① What is Linux ?

→ Linux is a kernel.

② what is an operating system ?

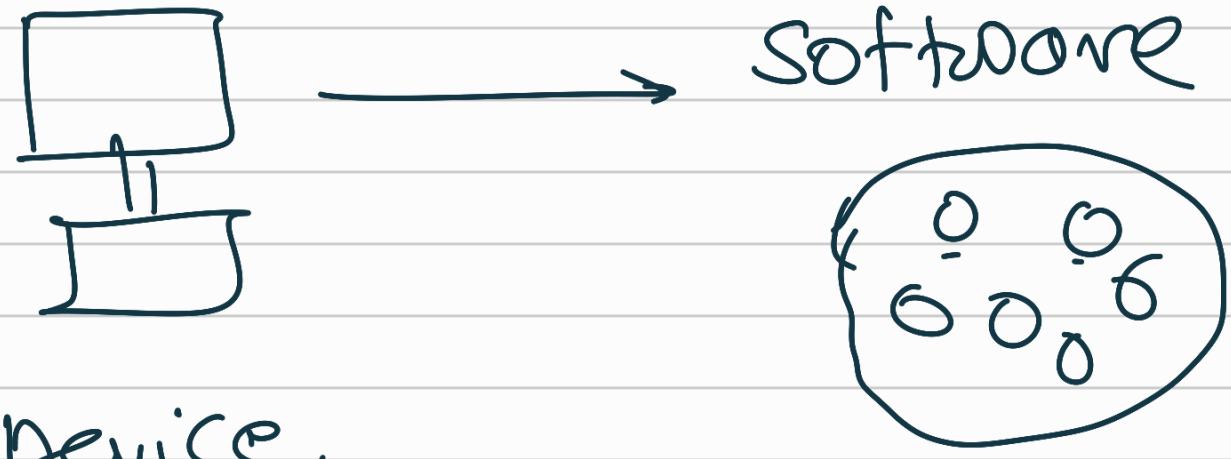
System - Device, laptop, desktop.

Operating / Software - To run this device we need software.

we create software through programming

- To run the device we need software, there are lots of programs exists.
- when there are lots of programs in one software & that software is capable to run the system, & that

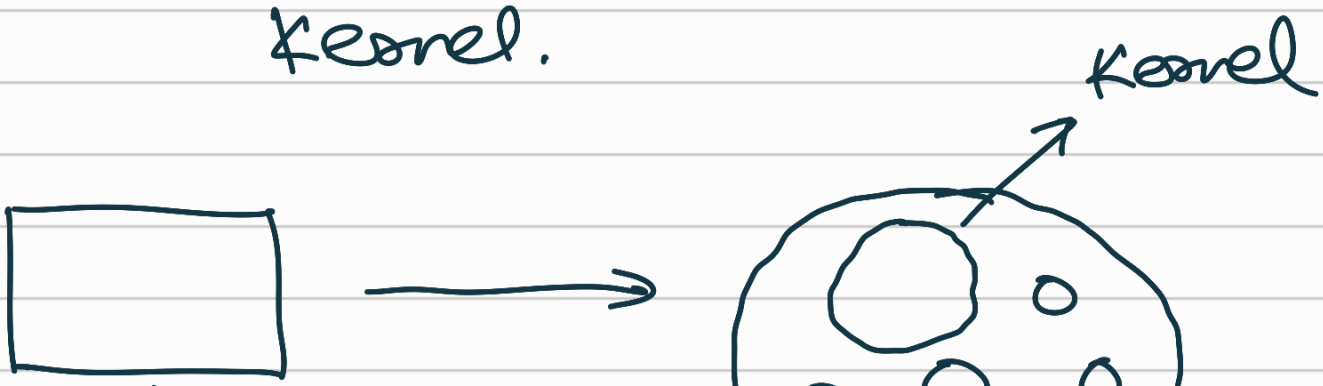
Software is known as operating system.

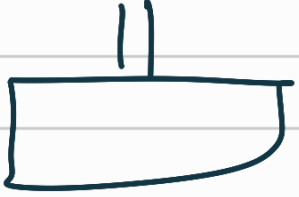


Device, laptop, desktop

⑤ What is kernel ?

There are lots of programs in the software, but there is one program who manages all other programs & that program is known as kernel.





Device, laptop, Desktop

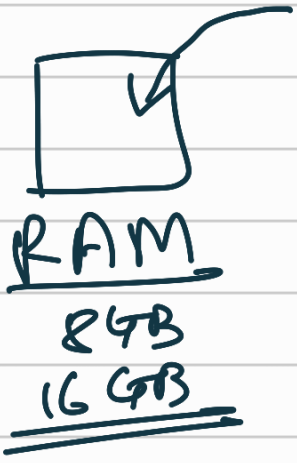


Operating/Software system

→ Kernel is the heart of the operating system

④ Job of Kernel :-

→ allocation of the RAM to the application





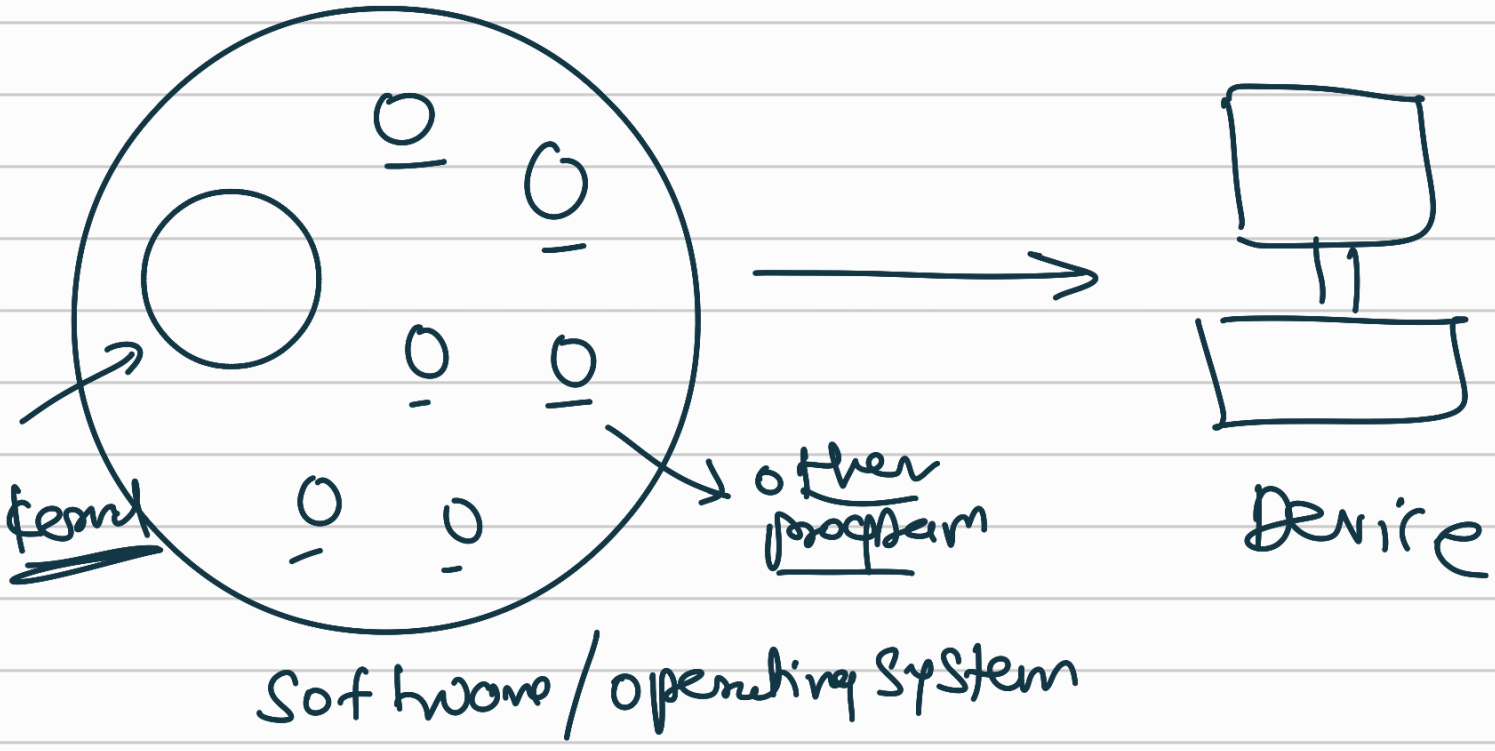
Disk

ROM (HDD, SDD)

→ allocation of CPU.

→ Linux Torvals developed Linux in 1991

→ Linux is a open source



Ubuntu — Ubuntu used this linux kernel. & wrote some programme & integrated to kernel & created a new software & made an operating system like ubuntu

① Ubuntu - OS
② fedora - OS
③ kali linux - OS

Using linux kernel

④ Red hat – OS

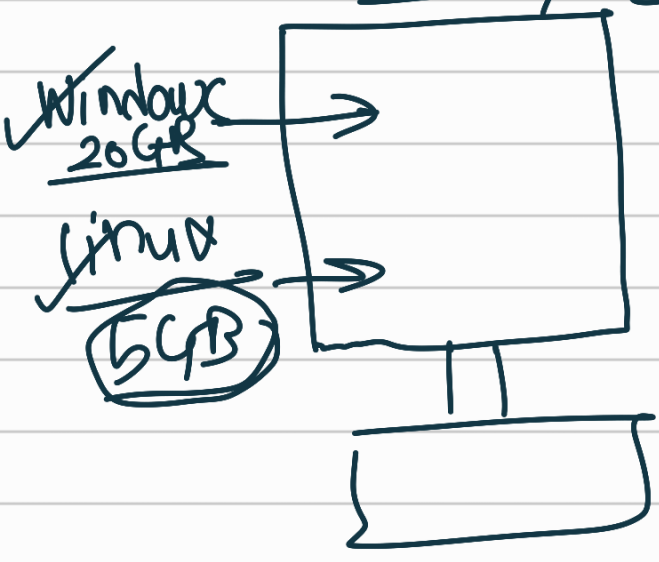
⑤ Amazon linux – OS

⑥ Centos- OS

- Features of linux

① Linux is light-weight OS

8GB / 2GB → 6GB



RAM – 8, 16, 32, 64

Processor – i3, i5, i7

ROM – 256GB, 512 GB

(HDD)

SSD

② Linux is secure OS

③ Linux provides
- Graphical based OS
- Command line based

Linux command syntax :-

Command means order to the OS.

eg. create folder
install application
uninstall application
delete folder
create .py file
delete .py file
copy file / folder
move file / folder
rename file / folder

Syntax :

Command option argument

ls -l /Desktop

Command - main command

option - required o/p as per our requirement

argument - apply on particular directory / location / folder.

1] pwd - Present working directory

It returns / prints current folder path / directory path.

$ pwd

→ /User/prashant

2] cd - change directory

$ cd /Desktop

$ cd /Users/prashant/logs

$ cd .. → previous directory

$ cd - → home directory

3] ls command - we get list of all files & folders.

$ ls → list of all files & folders

$ ls -l → details - owner, size, permission, etc

4] mkdir command -

$ pwd ——

$ mkdir linux

5] Vim Command

$ Vim demo.py

Step1 - press i → Insert

Step2 - write the content

Step3 - press esc

Step4 - :wq!

5] Cat Command - To open the file & read the content

$ Cat demo.py

6] rm - To remove file

$ rm file1.txt

rm -r - To remove folder

$ rm -r demo

7] rename

$ rename old_filename new_filename
old_filename

$ sudo apt update
$ sudo apt install rename

$ mv demo.py demo1.py

8] head & tail command

$ head -3 file1.py
o/p - top 3 lines

$ tail -3 file1.py

o/p - bottom 3 lines